



Cape Peninsula
University of Technology

FACULTY OF INFORMATICS & DESIGN

Diploma in Information & Communication Technology

APPLICATIONS DEVELOPMENT PRACTICE III: ADP372S

SUBJECT GUIDE:	2022
COURSE CODE:	ADP372S
NQF LEVEL:	7
PREREQUISITE SUBJECTS:	PREREQUISITE SUBJECTS: COURSE CODE: ADP262S NQF LEVEL: 6

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1. INTRODUCTION

Welcome to 3rd year Diploma in Information and Communication Technology. This subject is technology centric, designed to teach you essential enterprise application development skills, coupling the most current programming techniques with sound industry practices. The course builds on Applications Development Practice 2 and feeds off Applications Development Theory 3 as a guide in the structure of the work in this subject.

Java Programming language will be used as a programming vehicle to deliver this course. Please note that this course could also be delivered using C#, Python, Ruby or even PHP.

The course is being delivered in Java because, in first and second year you picked up the necessary Java skills. Note that this is NOT an Advanced Java Programming course, but rather a Software Development Application course.

2. GENERAL

2.1 Contact Information

DESIGNATION	NAME	LOCATION	TELEPHONE NUMBER	E-MAIL ADDRESS
Subject Co-ordinator	K. Naidoo	Engineering 2.11	(021) 460-3274	naidook@cput.ac.za
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Support Technician	R. Pietersen	Engineering 1.4	(021) 460-4209	pietersenr@cput.ac.za

2.2 Class Times and Venues

As published on <http://myclassroom.cput.ac.za> and notice boards. Due to the Covid-19 pandemic, Term 1 will continue with online classes and remote delivery. This may change as directed by the Department/Institution.

2.3 Learner Management System (LMS)

<https://myclassroom.cput.ac.za/>: Blackboard used for general communications and announcements, posting of course material, and assignments. It is the responsibility of the learner to check Blackboard regularly.

3. STUDY MATERIALS

3.1 Prescribed Textbook

No particular book is prescribed due to the technical nature of the contents taught, however learners will be directed to resources and be given material to use.

3.2 Recommended Readings

1. Pro Spring 5 Jan Machacek (Author), Aleksa Vukotic (Author), Anir-van Chakraborty (Author), Jessica Ditt (Author) ISBN-10: 1430218452 ISBN-13: 978-1430218456
2. Design Patterns: Elements of Reusable Object-Oriented Software [Hard-cover] Erich Gamma (Author), Richard Helm (Author), Ralph Johnson (Author), John Vlissides (Author) ISBN-10: 0201633612 ISBN-13: 978-0201633610
3. Test Driven Development: By Example Kent Beck (Author) ISBN-10: 0321146530 ISBN-13: 978-0321146533
4. Refactoring: Improving the Design of Existing Code Martin Fowler (Author), Kent Beck (Author), John Brant (Author), William Opdyke (Author), Don Roberts (Author) ISBN-10: 0201485672 ISBN-13: 978-0201485677
5. Java How to Program, 6/7/8th..10th Edition Harvey M. Deitel (Author), Paul J. Deitel (Author) ISBN-10: 0132222205 ISBN-13: 978-0132222204

3.3 Websites

- <https://docs.spring.io/spring-boot/docs/current/reference/htmlsingle/>
- <https://books.goalkicker.com/GitBook/>
- <https://git-scm.com/book/en/v2>
- <https://books.sonatype.com/mvnref-book/reference/index.html>
- <https://mvnrepository.com/>

4. MODULE CREDITS

This module has 20 SAQA or 0.167 HEMIS credits on NQF level 7. Please refer to the Faculty prospectus on <http://www.cput.ac.za/academic/faculties/informaticsdesign/prospectus> for further information.

4.1 Planning of Time Allocation for Learning Activities

There are 200 notional hours allocated to this module. These notional hours comprise of 60 hours (30%) for contact time and the remainder will be self-study and other tasks, assignments, and tests. *Due to the current Covid-19 pandemic, a multi-modal approach will be followed. The safety of the staff and the students are paramount as we navigate through this period.*

METHOD	% OF YOUR LEARNING TIME
Seminars / Practical's (face-to-face, limited interaction or technologically mediated)	30%
Independent self-study of standard texts and references and specially prepared materials (study guides, books, journals, using online tutorials etc.)	70%

5. ETHICAL CONDUCT

5.1 Roles and Responsibilities

Kindly refer to CPUT's Policy Docs and Student Guidelines:

https://www.cput.ac.za/storage/study_at_cput/2020/Student_Rules_Regulations_2020.pdf

5.2 Class attendance and participation

Kindly refer to page 18 of CPUT's General Handbook Academic and Student Rules and Regulations:
https://www.cput.ac.za/storage/study_at_cput/2020/Student_Rules_Regulations_2020.pdf

5.3 Grievance procedures

Kindly refer to page 61 of CPUT's General Handbook Academic and Student Rules and Regulations:
https://www.cput.ac.za/storage/study_at_cput/2020/Student_Rules_Regulations_2020.pdf

5.4 Plagiarism

Kindly refer to page 30 of CPUT's General Handbook Academic and Student Rules and Regulations:
https://www.cput.ac.za/storage/study_at_cput/2020/Student_Rules_Regulations_2020.pdf

**NB: An amended version of the General Handbook will be available on the CPUT website during the first/second term.*

6. ASSESSMENT

6.1 Assessment policy and regulations

This subject has two written assessments, practical assignments at the end of each theme, a practical exam and a capstone project where the knowledge and practical skills of the course will be integrated.

Test queries: Last day for any queries regarding a test is one week after the test was discussed in class. Thereafter the test mark cannot be changed.

Absence from a test: If you are absent from a test, a valid medical certificate must be given to your lecturer (via the Secretary) within one week after the test was written. You must also inform your lecturer on the day of the test that you are ill.

Due dates must be strictly adhered to. **NO late submissions are allowed for any subject in this Diploma.**

Assignments may be administered via Blackboard and other platforms.

6.2 Assessment Opportunities: Administration

This subject has 5 assessment weights namely T1, T2, T3, T4 and T5. T1 has a 10% weight and is an Assignment given in Term 1. T2 has a 30% weight, which will be a Practical Test in Term 2. T3 has 15% weight which is a Written Test at the end of Term 3. T4 has a 30% weight which is a Written Test administered in Term 4. Finally, T5 has a weight of 15%, derived from a Capstone Project that you will work on beginning in second term and based on the unique domain that you will need to obtain approval from the lecturer. The Capstone Project requires your integration of the knowledge and skills that you have acquired throughout the subject. Assessments may be handed in throughout the year for the Capstone Project. The Table below summarises the marks weighting.

		ASSESSMENT BREAKDOWN AND WEIGHTS				
ASSESSMENT CRITERIA	1 ST TERM	2 ND TERM	3 RD TERM	4 TH TERM		TOTALS
Nature of assessment	Assignment (T1)	Practical Test (T2)	Written Test (T3)	Written Test (T4)	Capstone Project (T5)	
Will assessment be included in this term?	Yes	Yes	Yes	Yes	Yes	100%
Total assessment weight for the term	10%	30%	15%	30%	15%	
Will the assessment be moderated?	No	Yes	No	Yes	No	
*This is subject to change as we return to contact classes on campus later in the semester						

STUDY COMPONENT

7. SUBJECT SPECIFICATIONS

7.1 Purpose of the Subject

This subject will equip you with the skills to design and implement software that not only correctly model business' need but also performs at scale using the most advanced software architectures. The strategic development skills will help you make best software design choices and integration decisions for any business requirements, resulting in applications and software models that explicitly reflect its core competencies. The tactical development skills will help you design useful software that accurately model unique operations of businesses. The subject will also equip you with knowledge about the broad options available to deploy solutions in a variety of infrastructures, whether on bare metal or in the cloud.

7.2 Subject Structure

7.2.1 Subject Outcomes

SUBJECT OUTCOMES	DESCRIPTION
SO1	Students will be able to prepare the Software Development Infrastructure with necessary tools for development and management of large software projects.
SO2	Integrate appropriate version control systems in an Integrated Development Environment to enable development of large-scale Projects
SO3	Integrate Appropriate Software Build Tools in an Integrated Development Environment to enable development of large-scale Projects
SO4	Students should understand test-driven development methodologies as part of the process of application development to increase application flexibility.
SO5	Students will be able to design effective, maintainable, object-oriented systems that exhibit low coupling or dependencies and high cohesion, the degree of relatedness
SO6	Students will be able to demonstrate how to use common Design Patterns to address software design issues, identify the most suitable design pattern to address a given design problem and critique code by identifying and refactoring anti-patterns.
SO7	Implementation of the Spring framework to reduce overall development time and increase efficiency by having a default setup for unit and integration tests. Furthermore to access and persist data between Java object/ class and the relational database.
SO8	Students will be able to apply application layer security constraints in software applications to make sure that only the right people can do only the right things when working through the application, including identity management, secure connections and secured data.
SO9	Demonstrate an understanding of Hexagonal Architecture and Application Deployment

7.2.2 Subject Breakdown Schedule

TERM 1				
STUDY UNIT 1: SOFTWARE DEVELOPMENT INFRASTRUCTURE AND DESIGN				
SUBJECT OUTCOMES	THEMES/SPECIFIC OUTCOMES	WEEK	ACADEMIC SCHEDULE	ASSESSMENT DATES
SO 1 SO 2 SO 3 SO 4	THEME 1: INTEGRATED DEVELOPMENT ENVIRONMENTS			
	SPO1: Overview of IntelliJ. SPO2: Creating a new project using the reverse domain name convention.	1	31 Jan – 18 Mar	
	THEME 2: VERSION CONTROL SYSTEMS			
	SPO3: Introduction to Git using the terminal SPO4: Using Git Tool within the IDE SPO5: Set up remote repository on Github with privileges.			
	THEME 3: BUILD TOOLS			
	SPO6: Using Maven with IntelliJ			
	THEME 4: TEST DRIVEN DEVELOPMENT			
	SPO7: Overview of Test Driven Development SPO8: Unit Testing			
	THEME 2: APPLICATION SOFTWARE DESIGN PRINCIPLES			
	SPO3: A recap/refresher of the fundamental coding principles of Java such as concrete & abstract classes, Interfaces and data structure.			
	SPO2: Student will be able to apply Software Design Principles to produce artefacts that have good dependencies			
	SPO3: Students will be able to apply Software Packaging Principles to produce designs that have high degree of relatedness and low dependencies.			
	THEME 3: SOFTWARE DESIGN PATTERNS IN PRACTICE			

	SPO1: Students will be able to use Creational Design patterns in object creation mechanisms, create objects in a manner suitable to the situation, to reuse and changeability.			
	SPO2: Students will be able to use Structure Patterns to ease the design of applications by identifying simple ways to realize relationships between entities in order to form large structures.			
	SPO3: Students will be able to use and apply Behaviour design patterns to solve common communication problems between objects and realize increase flexibility in carrying out this communication			
TEST WEEK				
RECESS				
PUBLIC HOLIDAYS AND OTHER IMPORTANT DATES: 14 Feb – Valentine’s Day 21 Mar – Human Rights Day				

TERM 2				
STUDY UNIT 2: DOMAIN DRIVEN DESIGN				
SUBJECT OUTCOMES	THEMES/SPECIFIC OUTCOMES	WEEK	ACADEMIC SCHEDULE	ASSESSMENT DATES
SO4 SO5 SO6 SO7	THEME 1: PRINCIPLES AND PRACTICES			
	SPO1: Using Domain Driven Design Concepts, students will be able to understand that a developer needs to value domain knowledge just as much as technical expertise to gain a deeper insight into the problem domain and to decompose large domains into smaller subdomains		28 Mar – 27 Jun	
	SPO2: Students will be able to create a common Ubiquitous Language and produce Context Mapping and Bounded Contexts for the problem space			
	SPO3: Given a software Requirements, students will be able to produce Domain Model Artefacts, such as class diagram			

	THEME 2: THE TACTICAL PATTERNS: EFFECTIVE DOMAIN MODELS		
	SPO1: Understand and apply a number of building block patterns to enable the creation of effective domain Model Objects such as Value Objects, Entities and Aggregate Routes		
	SPO2: Understand and apply a number of building block patterns to enable the creation of Abstractions such as Factories, Repositories and Services		
	SPO3: Understand and apply a number of building block patterns to enable the creation of Actions such as Commands, Events and Queries		
	THEME 3: STRATEGIC PATTERNS: BOUNDED CONTEXT COMMUNICATION		
	SPO1: Design Services to promote Bounded Context Integration	7	
	SPO2: Design Services to promote Messaging Integration and HTTP Integration through RPC and REST	8	
TEST WEEKS – 13 Jun to 25 Jun			
VACATION			
PUBLIC HOLIDAYS AND OTHER IMPORTANT DATES: • 15 Apr – Good Friday • 18 Apr – Family Day • 27 Apr – Freedom Day • 02 May 2022 Worker's Day (Observed) • 16 Jun – Youth Day • 17 Jun – University Holiday			

TERM 3

STUDY UNIT 3: APPLICATION SECURITY

SUBJECT OUTCOMES	THEMES/SPECIFIC OUTCOMES	WEEK	ACADEMIC SCHEDULE	ASSESSMENT DATES
SO8	THEME 1: SECURING APPLICATION			
	SPO1: Students will be able to analyse applications and gauge the security level of software applications based on Characteristics of Application Security	1	18 Jul – 09 Sep	
	THEME 2: AUTHENTICATION MECHANISMS			
	SPO1: Students will be able to choose the best Authentication Mechanism based on Basic Authentication, Form-Based Authentication	2		
	SPO2: Student will be able to apply HTTPS Client Authentication, Digest Authentication to applications	3		
	THEME 3: Case Study: Spring Framework Security			
	SPO1: Students will be design and implement secure HTTP/HTTPS Channel Security in an application	4		
	SPO3: Students will be able to choose and apply the Concurrent Session Control based on the needs of the application	5		
	SPO4: Students will be able to design application that deals with Session Fixation Attack Protection	6		
	SPO5: Student will able to Obtain information about the current user, Method Security in an application during runtime	7		
TEST WEEK				
VACATION				
PUBLIC HOLIDAYS AND OTHER IMPORTANT DATES:				
08 Aug – University Holiday 09 Aug – Women's Day				

TERM 4

STUDY UNIT 4: ARCHITECTURES AND DEPLOYMENT

SUBJECT OUTCOMES	THEMES/SPECIFIC OUTCOMES	WEEK	ACADEMIC SCHEDULE	ASSESSMENT DATES
SO9	THEME 1: ARCHITECTURES			
	SPO1: Hexagonal Architecture	1	19 Sep – 15 Dec	
	SPO2: Ports and Adapters	2		
	SPO3: Infrastructure, APIs and Domains	3		
	THEME 3: APPLICATION DEPLOYMENT AND API CONSUMPTION			
	SPO1: Packing and Deployment of Application in Production Environment	4		
	SPO2: Consumption of REST APIs with Web, Desktop and Mobile Phone Applications	5		
TEST WEEKS				
VACATION				
PUBLIC HOLIDAYS AND OTHER IMPORTANT DATES: • 24 Sep – Heritage Day • 16 Dec – Day of Reconciliation • 17 Dec – University Holiday • 25 Dec – Christmas Day • 26 Dec – Day of Good Will				

7.3 Articulation with other subjects in the programme

This subject is linked to concepts taught in Application Development Theory III

7.4 Learning presumed to be in place

The pre-requisite for this subject includes passing of all your majors at second year level

7.5 Critical Cross-field Outcomes

CRITICAL CROSS-FIELD OUTCOMES	DESCRIPTION
CCFO1	Identify and solve problems with responses which display responsible decisions, using critical thinking.
CCFO2	Work effectively with others as a member of a team, group, organisation or community.
CCFO3	Organise and manage oneself and one's activities responsibly and effectively.
CCFO4	Collect, analyse, organise and critically evaluate information.
CCFO5	Communicate effectively, using visual, mathematical, and/or language skills in the modes of written and/or oral presentation.
CCFO6	Use science and technology effectively and critically, showing responsibility towards the environment and the health of others.
CCFO7	Demonstrate an understanding of the world as a set of related systems by recognising that problem-solving contexts do not exist in isolation.
CCFO8	Contribute to the full personal development of each learner and the social and economic development of the society at large, by making it the underlying intention of any programme of learning to create awareness in individuals about the importance of: i) reflecting on and exploring a variety of strategies to learn more effectively; ii) participating as responsible citizens in the life of local, national and global communities; and iii) being culturally and aesthetically sensitive across a range of social contexts. (RSASAQA, 2001:24)

8. STUDY UNITS/STUDY THEMES

8.1 Study Units

Refer to 7.2.2 Subject Breakdown Structure.

8.2 Assessment Opportunities and Assessment strategy

Refer to 6.2 Assessment Opportunities: Administration for the linkages. Furthermore, a comprehensive rubric will be published on Blackboard for each assignment and project work.

8.3 Subject Outcomes and Associated Assessment Criteria

Refer to 7.2.2 Subject Breakdown Structure for the linkages. Furthermore, a comprehensive rubric will be published on Blackboard for each assignment and project work.

8.4 Self-study Activities

Use these links to deepen your knowledge

- <https://docs.spring.io/spring-boot/docs/current/reference/htmlsingle/>
- <https://books.goalkicker.com/GitBook/>
- <https://git-scm.com/book/en/v2>
- <https://books.sonatype.com/mvnref-book/reference/index.html>
- <https://mvnrepository.com/>

9. STUDENT ACKNOWLEDGEMENT

CAPE PENINSULA UNIVERSITY OF TECHNOLOGY
STUDENT ACKNOWLEDGEMENT
2022
DIPICTA: INFORMATION TECHNOLOGY
APPLICATION DEVELOPMENT PRACTICE III

ACKNOWLEDGE THAT YOU HAVE READ THE SUBJECT GUIDE AND UNDERSTAND ITS CONTENT

Please read the following statement and complete the spaces that follow. Make a copy of this page for your own reference. Hand-in this original page during the next class session.

I have read the subject guide of **APPLICATION DEVELOPMENT PRACTICE III** and understand the information and my responsibilities as a learner for this subject. I am also aware that the Subject Guide may be revised due to unforeseen circumstances and it remains my responsibility to adhere to the most recent guide.

Name & Surname:
(Print)

Student Number:

Contact number: (home/res) Contact number: (cell)

Signature: Date: